Design and implementation of an automatic nursing assessment system based on CDSS technology

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Abstract

BACKGROUND: Various quantitative and quality assessment tools are currently used in nursing to evaluate a patient's physiological, psychological, and socioeconomic status. The results play important roles in evaluating the efficiency of healthcare, improving the treatment plans, and lowing relevant clinical risks. However, the manual process of the assessment imposes a substantial burden and can lead to errors in digitalization. To fill these gaps, we proposed an automatic nursing assessment system based on clinical decision support system (CDSS). The framework underlying the CDSS included experts, evaluation criteria, and voting roles for selecting electronic assessment sheets over paper ones.

METHODS: We developed the framework based on an expert voting flow to choose electronic assessment sheets. The CDSS was constructed based on a nursing process workflow model. A multilayer architecture with independent modules was used. The performance of the proposed system was evaluated by comparing the adverse events' incidence and the average time for regular daily assessment before and after the implementation.

RESULTS: After implementation of the system, the adverse nursing events' incidence decreased significantly from 0.43 % to 0.37 % in the first year and further to 0.27 % in the second year (p-value: 0.04). Meanwhile, the median time for regular daily assessments further decreased from 63 s to 51 s.

CONCLUSIONS: The automatic assessment system helps to reduce nurses' workload and the incidence of adverse nursing events.

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